Gaining Manufacturing Quality Experience at John Deere

If you already have **strong experience as a product quality engineer** (what John Deere often calls PDP/Program Quality Engineer or similar new-product-focused roles), spending time as a Manufacturing Quality Engineer (or “Quality Engineer, Assembly” type role) is a common and valuable lateral move at Deere. It rounds you out in ways that pure product development experience often lacks.

What You Gain (Biggest Upsides)

- **Hands-on production process mastery** — You’ll live the day-to-day realities of how Deere actually builds equipment (welding, machining, paint, assembly lines, material flow). Product QE experience is mostly theoretical or early-stage; manufacturing QE teaches you real tolerances, variability, operator challenges, and why certain design features cause headaches on the line.
- **Real-time root cause and containment skills** — You’ll handle live quality escapes, high-impact deficiencies, inspections, and immediate corrective actions on the production floor. This sharpens your troubleshooting far beyond design-phase risk tools. Many product QEs have never truly owned a line stop or customer escape.

- **Operational excellence & continuous improvement muscle** — Heavy exposure to Kaizen, 5S, mistake-proofing (poka-yoke), scrap reduction, and the full Safety-Quality-Delivery-Cost (SQDC) mindset in a running factory. This is gold when you later circle back to product roles and push better Design for Manufacturability (DFM).
- **Stronger cross-functional influence with operations** — You build relationships with production supervisors, operators, and manufacturing engineers. This credibility is huge at Deere plants (Waterloo, Dubuque, Thibodaux, etc.) and makes you more effective in any future quality leadership position.
- **Broader career optionality** — Deere values people who've "been in the factory." It opens doors to roles in Supplier Quality, Plant Quality Management, or even back into programs with real production insight.

In short: You become the quality engineer who actually understands ***how things are built****, not just how they're designed.*

What You Lose (or Get Significantly Less Of)

- **New-product development depth (EPDP lifecycle)** — Much less time on early-phase quality planning, system/design FMEAs, and new program launches. PDP roles keep you embedded with Product Line Management Teams and design engineers from concept through validation; manufacturing roles are mostly post-launch sustainment.
- **Preventive quality tools at the design level** — You'll do more PFMEA/Control Plan work on the floor but far less DFMEA facilitation and upfront risk mitigation for brand-new products. The preventive/strategic side slows down.
- **Strategic program representation** — Less visibility representing Quality on cross-functional product development teams and less influence on future product

roadmaps. Manufacturing QE is more embedded with Factory/Operations leadership than with R&D.

- **Exposure to innovation and cutting-edge features** — You'll work on current production (sometimes older platforms) rather than the newest tractor, sprayer, or construction equipment programs. The "exciting new product" energy decreases.
- **Potentially slower path to senior program or enterprise quality roles** — If your long-term goal is PDP Quality Manager, Quality Director on major platforms, or advanced engineering, staying purely manufacturing-focused for too long (3+ years) can make you less visible on the development side.

Bottom Line Recommendation

Most people at John Deere who make this move for **1–3 years** come back to product/program quality (or get promoted) as much stronger engineers because they can now speak the language of both design **and** production. It's especially powerful if you want to move into Plant Quality Manager, Supplier Development, or cross-functional leadership later.

The "loss" is mainly temporary — Deere has excellent internal mobility, and many Quality Engineers rotate between PDP and manufacturing roles over their career.

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